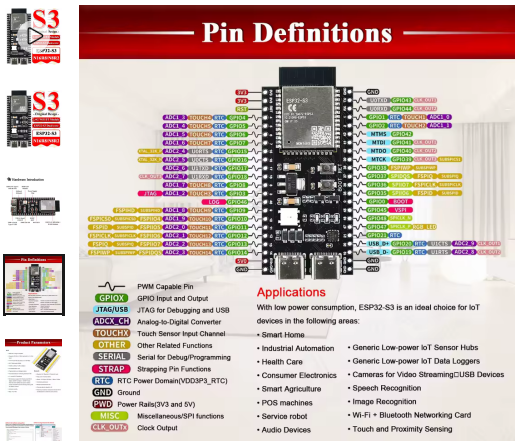
€ Import charges will apply to non-EU shipments



ESP32-S3 Wifi BT Module Development Board for Arduino IDE ESP32-S3 N16R8 N8R2 44Pin Type-C 16MB Flash 8M PSRAM ESP32 S3

4.9 2297 Reviews | 10,000+ sold

11.11 Sale Starts: Nov 1, 00:00 CET

7,89€ Upcoming price 7,69€

7,49€
≥ 2 pieces

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Color: S3-N8R2 with Cable

AI overview of item

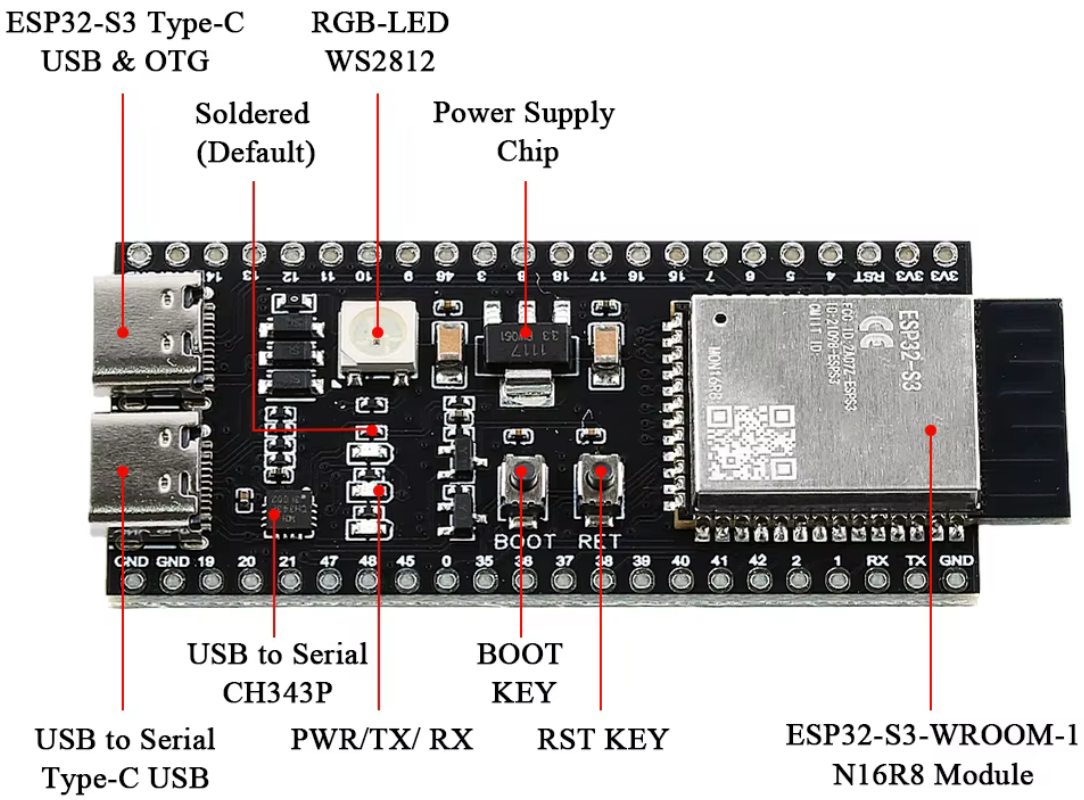
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- **Powerful ESP32-S3 Development Module** Equipped with the **ESP32-S3-WROOM1** chip, this module delivers high performance for IoT and wireless applications, supporting both **Wi-Fi** and **Bluetooth** connectivity with advanced processing capabilities.
- **High Memory Capacity for Complex Tasks** Featuring **16MB Flash** and **8MB PSRAM**, this module supports large firmware and data storage, ideal for running complex applications and multitasking in embedded systems.
- **Compatible with Arduino IDE** Fully compatible with the **Arduino IDE**, this board enables easy programming and development for hobbyists and professionals alike, simplifying the integration of wireless features into projects.
- **44-Pin Type-C Design for Easy Connection** The **44-pin Type-C** interface provides a stable and convenient connection, supporting fast data transfer and power delivery, making it ideal for prototyping and development workflows.
- **Supports Multiple Models and Variants** Available in **S3-N16R8** and **S3-N8R2** versions, this module offers flexibility for different project needs, with options including or excluding a cable for varied setup preferences.
- **Standard Operating Conditions for Reliable Use** Designed for **standard operating temperature** and **standard dissipation power**, this module ensures stable performance in typical indoor environments and general electronic applications.
- **Brand-Free, New, and Ready to Use** As a **new** and **brand-free** module, it provides a cost-effective solution for developers seeking a reliable, high-performance board without additional branding overhead.

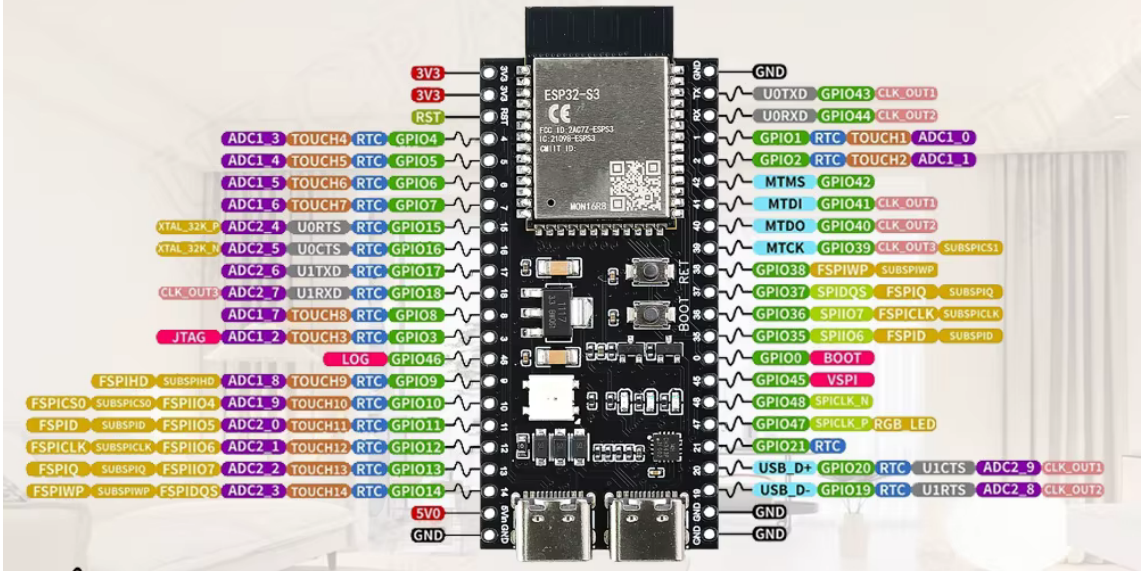
view less


ESP32-S3
FCC ID:2AC7Z-ESPS3
IC:21098-ESPS3
CMIIT ID:
MON16R8
BOOT
RST
S3
- Original Design -
2.4G Wifi BT Module
ESP32-S3 Dual-core
ESP32-S3
N16R8/N8R2

Hardware Introduction



Pin Definitions



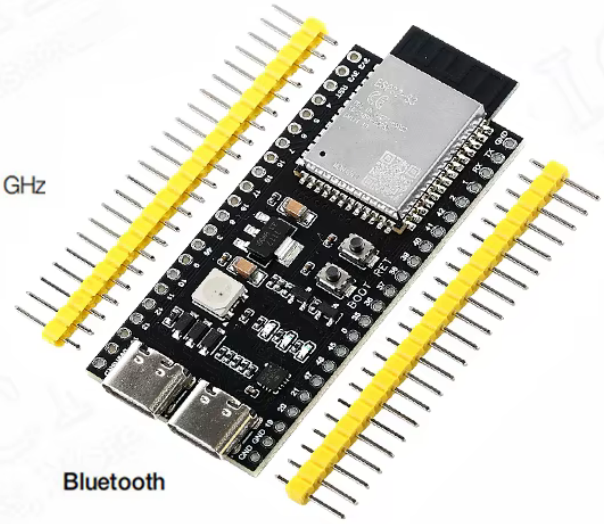
PWM Capable Pin
GPIOX GPIO Input and Output
JTAG/USB JTAG for Debugging and USB
ADCX_CH Analog-to-Digital Converter
TOUCHX Touch Sensor Input Channel
OTHER Other Related Functions
SERIAL Serial for Debug/Programming
STRAP Strapping Pin Functions
RTC RTC Power Domain(VDD3P3_RTC)
GND Ground
PWD Power Rails(3V3 and 5V)
MISC Miscellaneous/SPI functions
CLK_OUTx Clock Output

Applications

With low power consumption, ESP32-S3 is an ideal choice for IoT devices in the following areas:

- Smart Home
- Industrial Automation
- Health Care
- Consumer Electronics
- Smart Agriculture
- POS machines
- Service robot
- Audio Devices
- Generic Low-power IoT Sensor Hubs
- Generic Low-power IoT Data Loggers
- Cameras for Video Streaming□USB Devices
- Speech Recognition
- Image Recognition
- Wi-Fi + Bluetooth Networking Card
- Touch and Proximity Sensing

Product Parameters



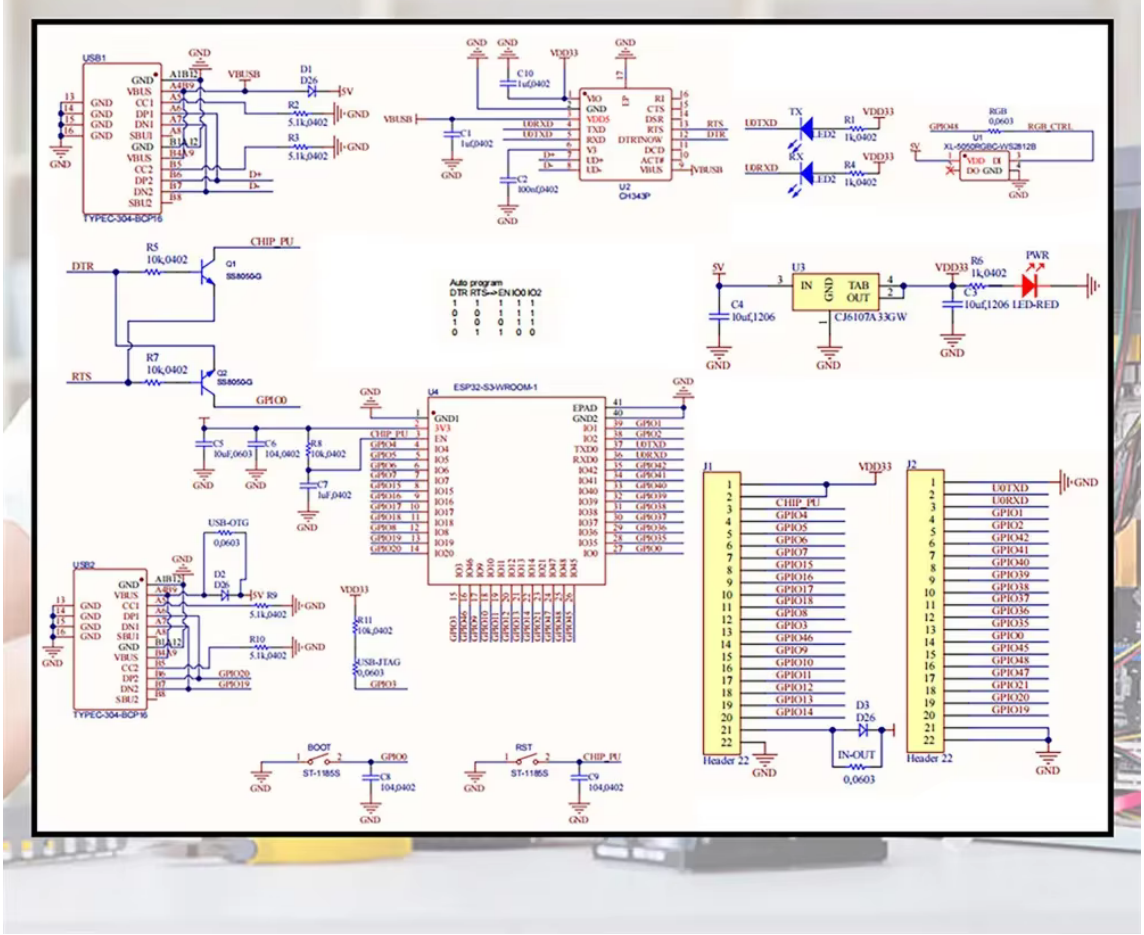
Wi-Fi

- IEEE 802.11b/g/n-compliant
- Supports 20 MHz, 40 MHz bandwidth in 2.4 GHz band
- 1T1R mode with data rate up to 150 Mbps
- Wi-Fi Multimedia (WMM)
- TX/RX A-MPDU, TX/RX A-MSDU
- Immediate Block ACK
- Fragmentation and defragmentation
- Automatic Beacon monitoring (hardware TSF)
- 4 × virtual Wi-Fi interfaces
- Simultaneous support for Infrastructure BSS in Station, SoftAP, or Station + SoftAP modes Note that when ESP32-S3 scans in Station mode, the SoftAP channel will change along with the Station channel
- Antenna diversity
- 802.11mc FTM

Bluetooth

- Bluetooth LE: Bluetootn 5, Bluetooth mesh
- High power mode (20 dBm)
- Speed: 125 Kbps, 500 Kbps, 1 Mbps, 2 Mbps
- Advertising extensions
- Multiple advertisement sets
- Channel selection algorithm #2
- Internal co-existence mechanism between Wi-Fi and Bluetooth to share the same antenna

Schematic


USB1
TYPEC-304-BCP16
D1
D26
5V
VBUSB
R2
5.1k,0402
R3
5.1k,0402
C10
1uf,0402
C1
1uf,0402
C2
100nf,0402
U2
CH343P
U0TXD
U0RXD
TX
RX
LED2
R1
1k,0402
R4
1k,0402
VDD33
RGB
0,0603
RGB_CTRL
U1
XL-5050RGBC-WS2812B
R5
10k,0402
CHIP_PU
Q1
S8050-G
DTR
R7
10k,0402
Q2
S8050-G
RTS
GPIO0
Auto program
DTR RTS→EN IO0 IO2
U3
IN
GND
TAB
OUT
CJ6107A 33GW
C4
10uf,1206
C3
10uf,1206 LED-RED
R6
1k,0402
PWR
U4
ESP32-S3-WROOM-1
C5
10uf,0603
C6
104,0402
R8
10k,0402
C7
1uf,0402
USB2
USB-OTG
0,0603
R9
5.1k,0402
R10
5.1k,0402
R11
10k,0402
USB-JTAG
0,0603
GPIO20
GPIO19
GPIO3
J1
J2
Header 22
D3
D26
IN-OUT
0,0603
BOOT
ST-11B5S
C8
104,0402
RST
ST-11B5S
C9
104,0402
GND
VDD33
5V

Note on the storage situation of the ESP32-S3 module

mpn	内置sram	内置rom	外扩psram	外扩flash
ESP32-S3-N4	512K	384K	0M	4M
ESP32-S3-N8	512K	384K	0M	8M
ESP32-S3-N16	512K	384K	0M	16M
ESP32-S3-N4R8	512K	384K	8M	4M
ESP32-S3-N4R2	512K	384K	2M	4M
ESP32-S3-N8R2	512K	384K	2M	8M
ESP32-S3-N16R2	512K	384K	2M	16M
ESP32-S3-N8R8	512K	384K	8M	8M
ESP32-S3-N16R8	512K	384K	8M	16M

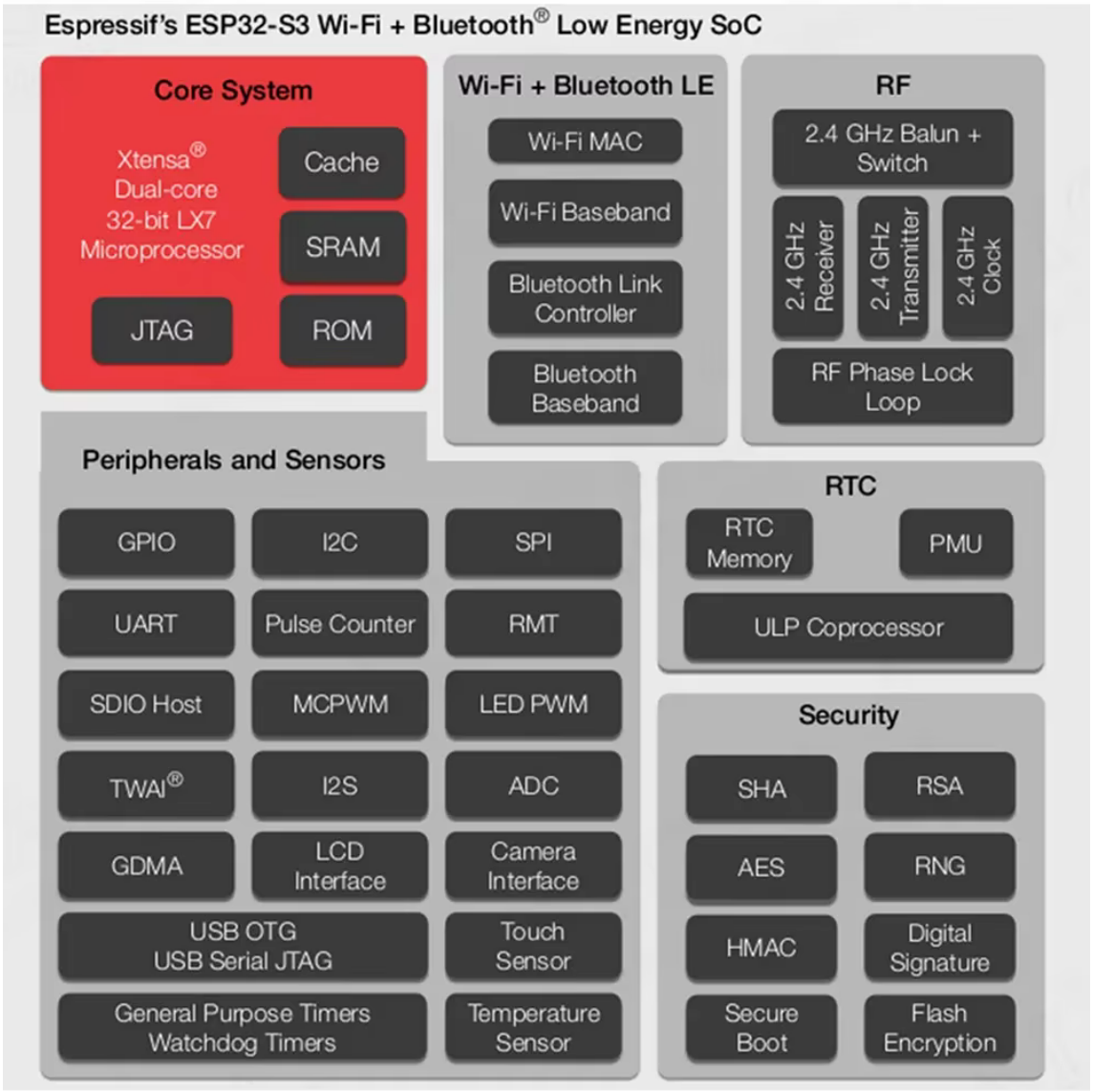
The link can choose N8R2/N8R8/R16R8 specifications of goods, the series in turn downward compatible, storage specifications of the module program can be stored without modification, directly run on the higher specifications of the storage module, for example, the N4 program can be run on top of the N8, N16, the specifications of the module program can be run on top of the expansion of the prom 0 expansion of the psram module of 2M, 8M.

About ESP32-53 hardware resources

ESP32-S3 is a low-power MCU SoC that integrates 24ghz Wi-Fi and Bluetooth low energy dual-mode wireless communications. ESP32-S3 has a full Wi-Fi and Bluetooth low energy subsystem, providing industry-leading low power and RF performance. Support a variety of low-power working state, can meet the power requirements of various application scenarios. ESP32-S3 chip provides rich peripheral interfaces, and has a variety of unique hardware security mechanisms. The Perfect Security Mechanism enables the chip to meet the strict security requirements.

Performance:

- **Core:**
  - **Xtensa® dual-core 32-bit LX7 CPU, frequency up to 240MHz**
- **Memories:**
  - **384 KB of ROM**
  - **512 KB of SRAM**
  - **16 KB of RTCSRAM**
  - **8 MB of PSRAM**
- **Working Voltage: 3 V to 3.6 V**
- **Up to 45 GPIOs**
- **2*12-bit ADC (up to 20 channels)**
- **Communication interfaces**
  - **2 I2C interfaces**
  - **2 I2S interface**
  - **4 SPI interfaces**
  - **3 UART interfaces**
  - **1 USB OTG interface**
- **Security:**
  - **4096 bit OTP**
  - **AES, SHA, RSA, ECC, RNG**
  - **Secure Boot, Flash Encryption, Digital signature, HMAC module**
- **Extended temperature range: -40 to 65 °C**


Espressif's ESP32-S3 Wi-Fi + Bluetooth® Low Energy SoC
Core System
Xtensa® Dual-core 32-bit LX7 Microprocessor
Cache
SRAM
ROM
JTAG
Wi-Fi + Bluetooth LE
Wi-Fi MAC
Wi-Fi Baseband
Bluetooth Link Controller
Bluetooth Baseband
RF
2.4 GHz Balun + Switch
2.4 GHz Receiver
2.4 GHz Transmitter
2.4 GHz Clock
RF Phase Lock Loop
Peripherals and Sensors
GPIO
I2C
SPI
UART
Pulse Counter
RMT
SDIO Host
MCPWM
LED PWM
TWAI®
I2S
ADC
GDMA
LCD Interface
Camera Interface
USB OTG
USB Serial JTAG
Touch Sensor
General Purpose Timers
Watchdog Timers
Temperature Sensor
RTC
RTC Memory
PMU
ULP Coprocessor
Security
SHA
RSA
AES
RNG
HMAC
Digital Signature
Secure Boot
Flash Encryption



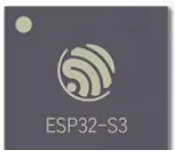
WiFi

- Support IEEE802.11b/g/n protocols
- Supports 20 MHZ and 40MHz bandwidth in 24GHZ band.
- Supports 1T1R mode with data rate up to 150Mbps.
- Wireless Multimedia (WMM)
- Frame Aggregation (TX/RXA-MPDUTX/RXA-MSDU)
- Immediate Block ACK (IBAC)
- Fragmentation and reorganization (Fragmentation/defragme ntation)
- Beacon automatic monitoring (hardware TSF)
- 4x Virtual Wi-Fi Interface
- Also supports Infrastructure BSS (Infrastructure BSS) Station Mode, SoftAP Mode and Station+SoftAP Mixed Mode, please note that when ESP32-S3 is scanning in Station Mode, the SoftAP channel will be changed at the same time.
- Antenna Diversity
- 802.11mcFTM
- Support external power amplifier

Bluetooth

- Low Power Bluetooth (Bluetooth LE): Extensions)Bluetooth 5, Bluetooth mesh
- High power mode (20 dBm, shared PA with Wi-Fi)
- Rate support 125 Kbps, 500Kbps, 1Mbps, 2Mbps Advertising Extensions
- Multiple Advertisement Sets
- Channel Selection (Channel Selection Algorithm #2)
- Wi-Fi and Bluetooth coexist and share the same antenna.





The ESP32-S3 is an integrated 2.4 GHz Wi-Fi and Bluetooth 5 (LE) MCU with Long Range support, powered by the Xtensa (R) 32-bit LX7 dual-core processor with 240 MHz, 512 KB of internal SRAM (TCM), 45 programmable GPIO pins and a rich set of ESP32-S3 supports higher capacity high-speed Octal

Feetech 1910 Servo
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or
Upload a photo
*For a quick search hit CTRL+V to paste an image into the search box

Customer Reviews (2297) Specifica e



ructions to accelerate
ignal processing.AI
structions through the
alize high-performance
ition, speech wake-up
-Skainet will also support
this feature. Skainet will also support this feature.



Wi-Fi + Bluetooth 5 (LE)

The ESP32-S3 integrates 2.4GHz Wi-Fi (802.11 b/g/n) supporting 40 MHz bandwidth; its low-power Bluetooth subsystem supports Bluetooth 5(LE) and Bluetooth Mesh for long-distance communication with broadcast extensions via Coded PHY. It also supports 2Mbps PHY for higher transmission speed and data throughput.The ESP32-S3 has excellent Wi-Fi and Bluetooth LE RF performance, and can work stably at high temperatures.



Sophisticated security mechanisms

The ESP32-S3 provides a complete security mechanism and protection for IoT devices against all kinds of malicious attacks and threats. It supports flash alarm based on AES-XTS algorithm, secure boot based on RSA algorithm, digital signature and HMAC. ESP32-S3 also adds a new "world controller" module, which provides two non-interfering execution environments and realizes the trusted execution environment or privilege separation mechanism. ESP32-S3 also adds a new "world controller" module, which provides two execution environments that do not interfere with each other, realizing a trusted execution environment or privilege separation mechanism.



Mature software support

ESP32-S3 is based on Loxin's mature IoT development framework, ESP-IDF, which has successfully empowered hundreds of millions of IoT devices, undergone rigorous testing and release cycles, and has a clear and effective support strategy. Based on its mature software architecture, developers will find it easier to build applications or migrate legacy programs to the ESP32-S3 platform, thanks to its familiarity with the tools and APIs.



Rich IO interfaces

The ESP32-S3 features 45 programmable GPIOs as well as popular peripheral interfaces such as SPI, I2S, I2C, PWM, RMT, ADC, UART, SD/MMC host controller and TWAITM controller. Fourteen of the GPIOs can be configured as capacitive touch inputs for HMI interaction. In addition, the ESP32-S3 is equippec with an ultra-low-power co-processor (ULP), which supports a variety of low-power modes and is widely used in various low-power application scenarios.

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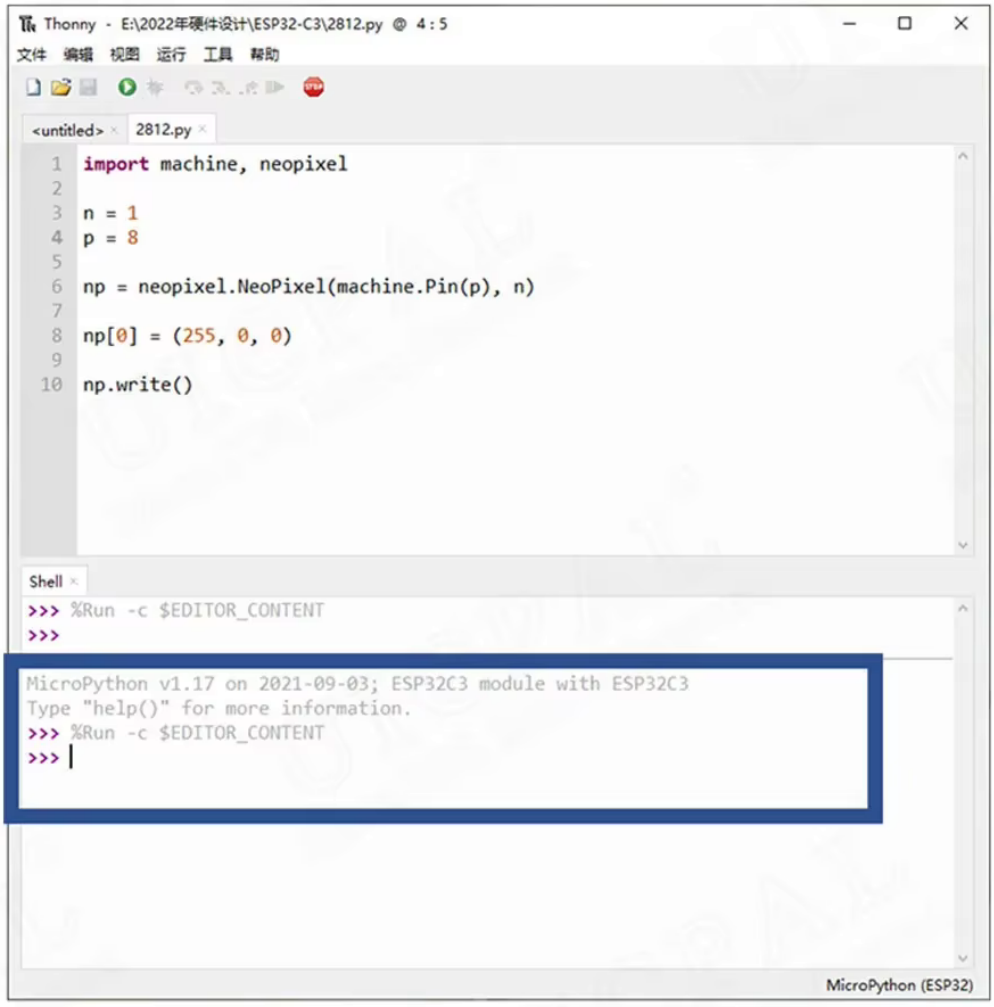
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Unavailable(4)

ESP32-S3 IDE software using python

ESP32-S3 can be programmed and controlled using Micropython for the Python language IDE is recommended to use the MIT-licensed Thonny software developed at the University of Estonia.



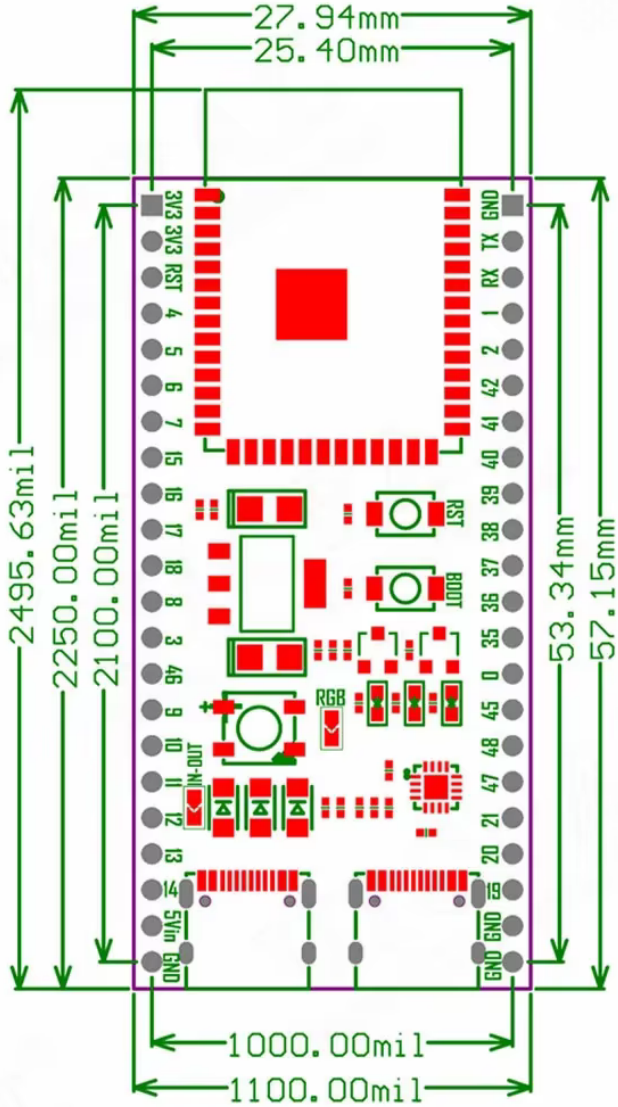
Low Power Management

- Power management unit with five power consumption modes
- Ultra-low power co-processor (ULP):
  -ULP-RISC-V coprocessor.
  -ULP-FSM coprocessor

Secure Mechanism

- Secure Boot
- Flash Encryption
- 4096-bit OTP, user-available up to 1652 bits
- Encryption hardware gas pedal
  - AES-128/256(FIPS PUB197)
  - Hash (FIPS PUB 180-4)
  -RSA
  - Random Number Generator (RNG)
  - HMAC
  - Digital Signature

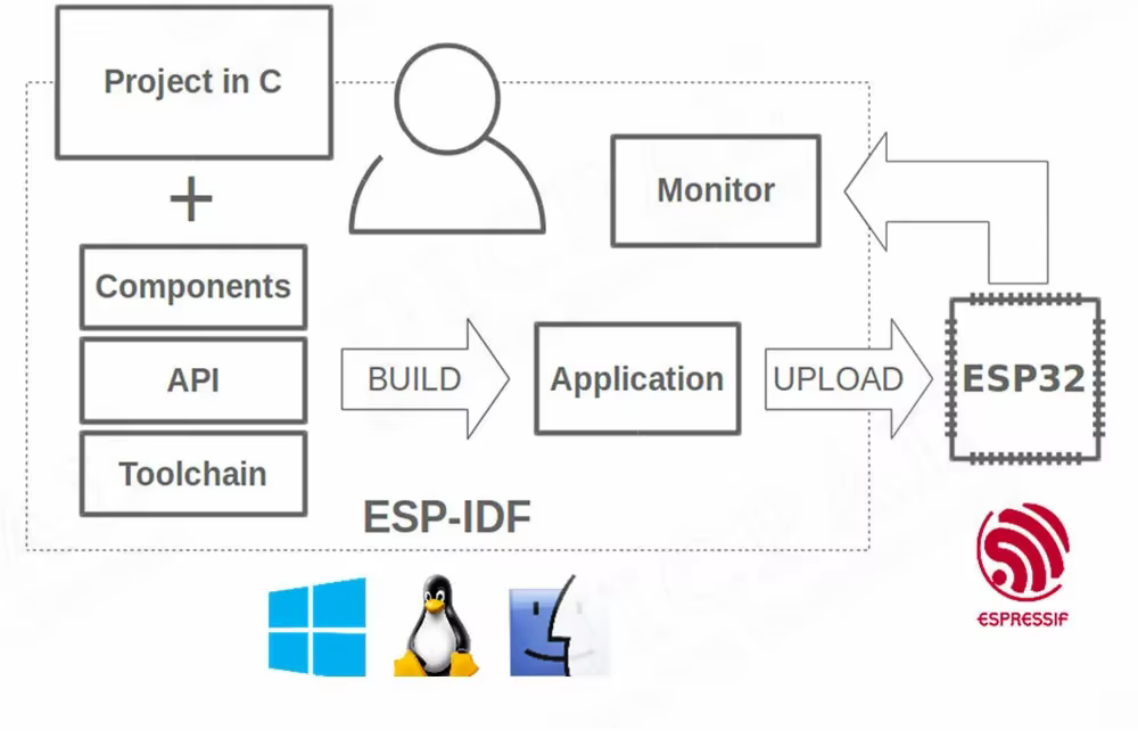
Dimension Drawing



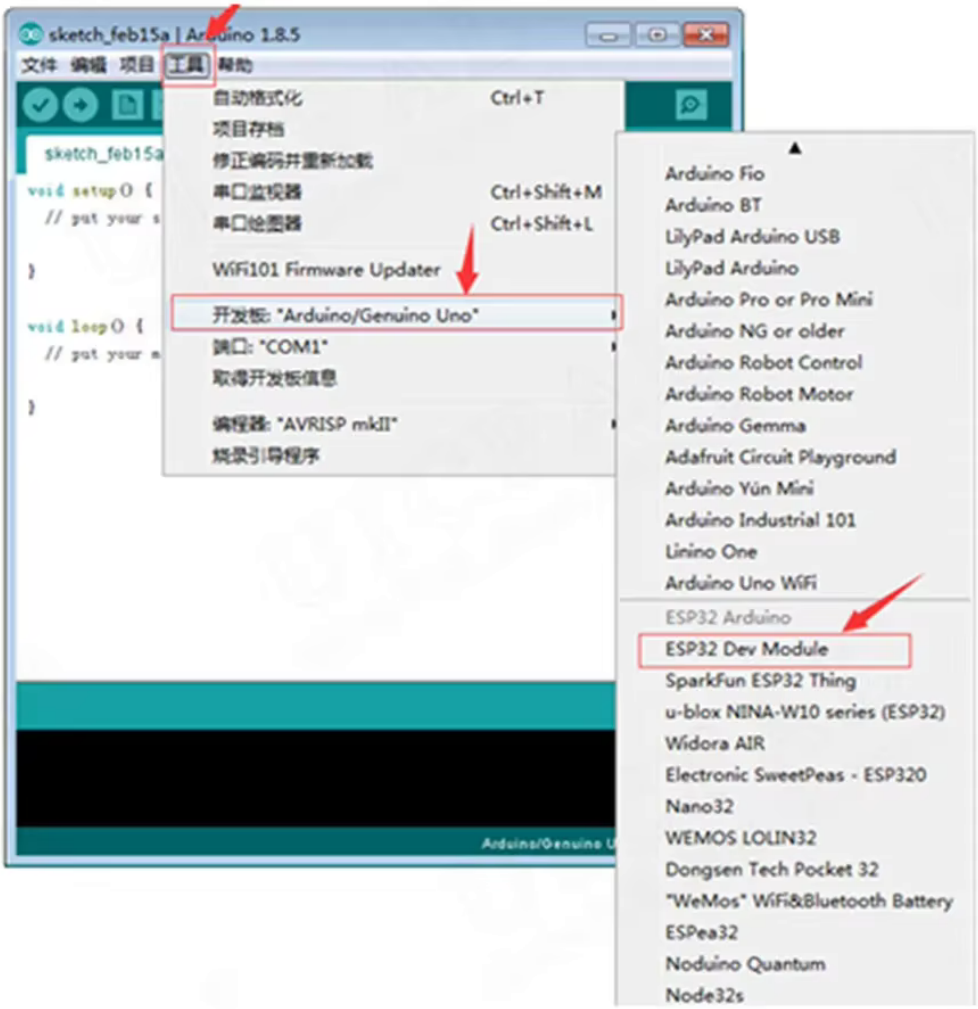
100mil = 2.54mm

ESP32-S3 Programming with IDF

ESP-IDF, Loxin IoT Development Framework, provides toolchain, API, component and workflow support for developing ESP32-S3 applications on Windows, Linux and macOS system platforms.

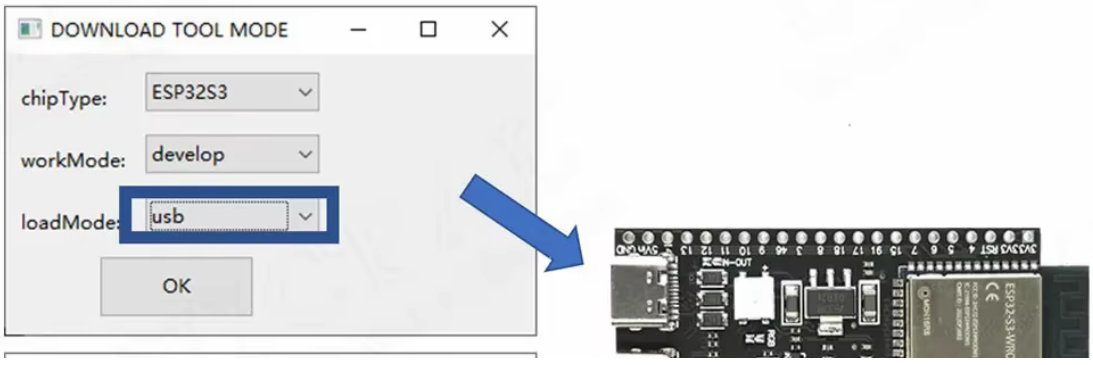


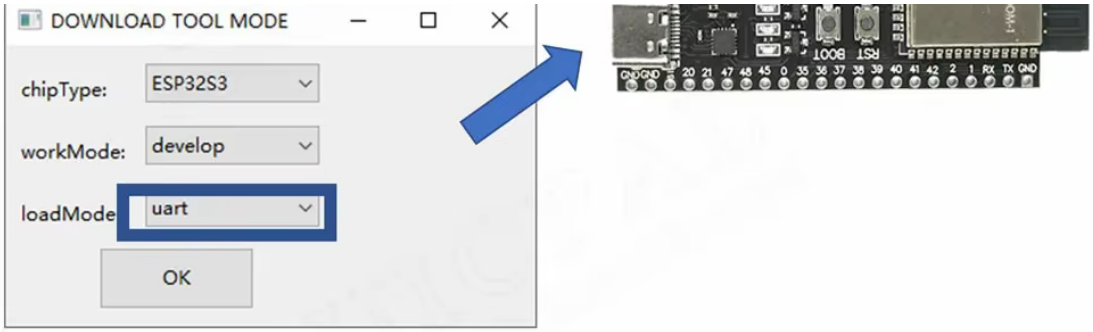
ESP32-S3 using Arduino's IDE software



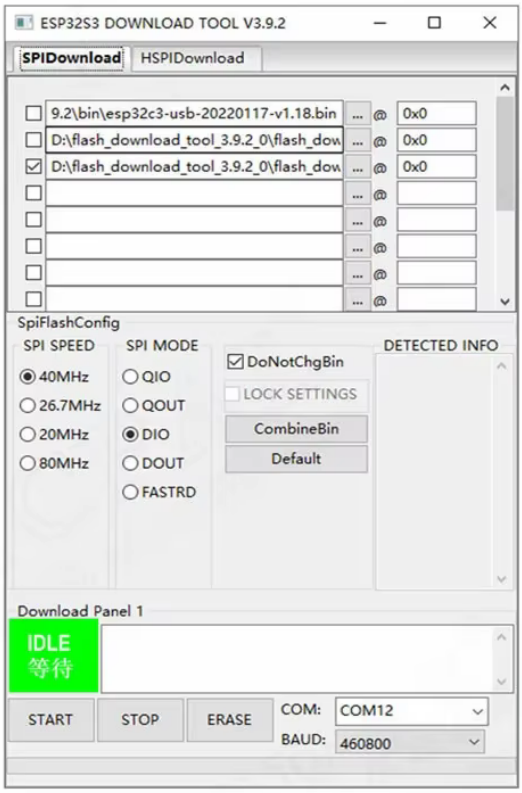
About ESP32-S3 how to Download

ESP32-S3 can download the burn-in program file (burn-in firmware) through the ESP32 pass-through USB interface, or through the on-board hardware USB-to-serial port, in short, the two TYPE-CUSB ports on the board can download the program. Under windows environment, you can download the program through the official flash download tool xxx software, note that the two USB port mode one is called USB mode, one is UART mode.





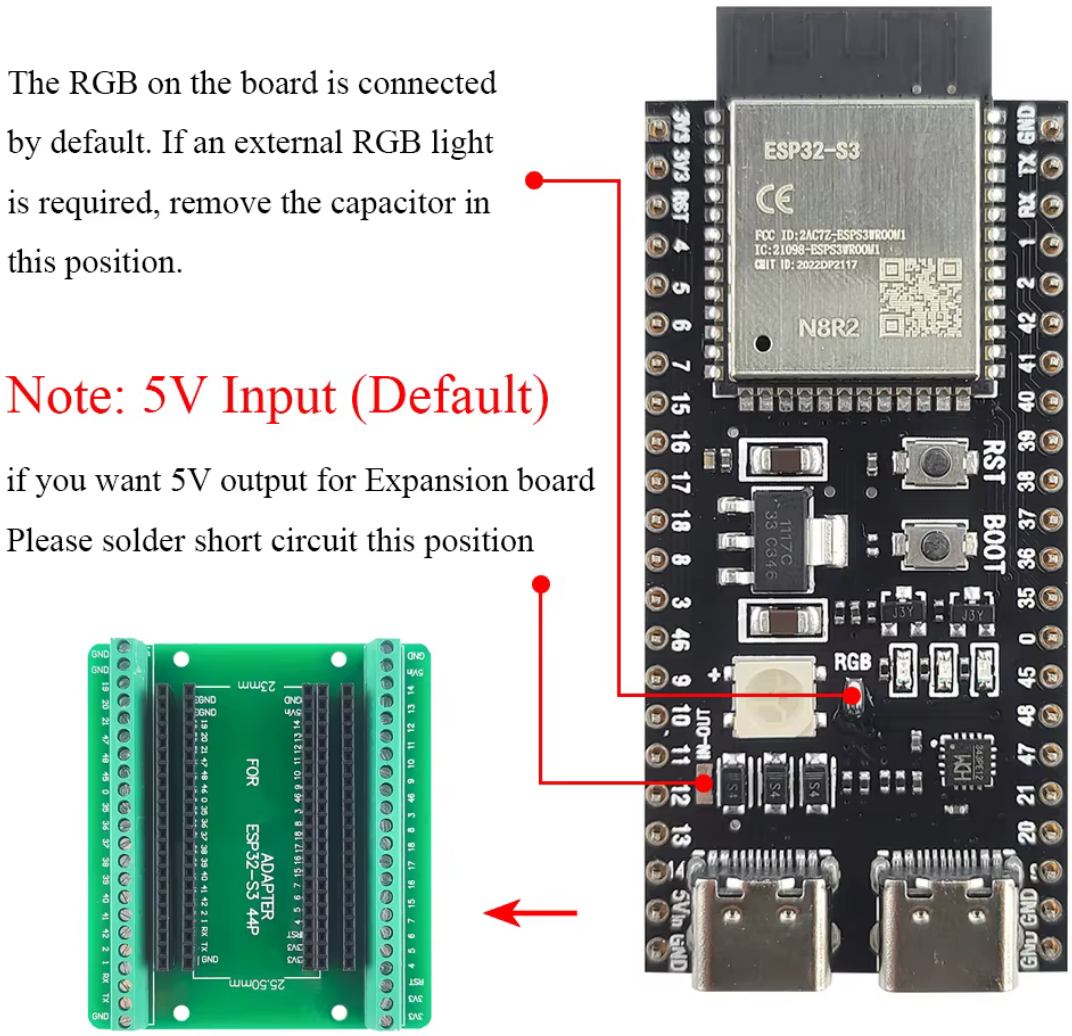
ESP32-S3 Download Burn File Software



Whether it is USB or serial mode download, you need to erase the operation before downloading. Then carry out the corresponding download operation Note that even if the USB download is to use the USB virtual serial port to download regardless of the core board in any case as long as it is powered on and running, by holding down the boot button, press 1 RST will be able to let the hardware system in standby download mode (booloard), waiting for the user to download the firmware.

Advanced Peripheral Interfaces and Sensors

- 45 x GPIO port
- Digital interfaces.
  - ◆-4xSPI
  - ◆-1xLCD interface (8-bit~16-bit parallel RGB, I8080, MOTO6800), YUV422, YUV420, YUV411 conversion between each other.
  - ◆-1xDVP 8-bit~16-bit camera interface
  - ◆- 3xUART
  - ◆- 2xI2C
  - ◆- 2×I2S
  - ◆- 1x RMT(TX/RX)
  - ◆- 1×Pulse Counter
  - ◆- LED PWM Controller, up to 8 channels
  - ◆-1×Full Speed USB OTG
  - ◆-1×USB Serial/JTAG Controller
  - ◆-2×MCPWM
  - ◆-1× SDIO host interface with 2 card slots
  - ◆- General Purpose DMA controller (GDMA for short), 5 receive channels and 5 transmit channels
  - ◆-1×TWA controller with ISO11898-1 (CAN specification 2.0)
- Analog Interface.
  - ◆-2 x 12-bit SAR ADC, up to 20 channels
  - ◆-1 x temperature sensor
  - ◆-14 x capacitive sensing GPIOs
- Timer: - 1X RMT(TX/RX)
  - ◆- 4×54-bit general-purpose timer
  - ◆-1x 52-bit System Timer
  - ◆-3×Watchdog Timer



Buyer Questions & Answers (12)

What gpio pin controls the rgb led ...

Could you release an announcement with options to choose more units per purchase? ...

What is this and what is it for? ...

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